

Measured RAG

Grounded answers. Inspectable retrieval. Reproducible evaluation.

A production-style reference implementation for document question answering that makes retrieval choices, evidence, diagnostics, and evaluation artifacts visible instead of hiding them behind a chat interface.

What the system demonstrates

Hybrid retrieval Dense retrieval and BM25 run for every route, followed by weighted fusion and cross-encoder reranking.	Evidence inspection Users can inspect indexed chunks, final retrieved sources, route choices, scores, and timing breakdowns.	Measured behavior A frozen QASPER-derived evaluation records answer overlap, semantic compatibility, latency, and provenance.
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Core workflow

01	Ingest	Extract text from PDF, DOC, or DOCX files and create configurable character-window chunks.
02	Retrieve	Run dense and lexical search, fuse their ranked candidates, then rerank with a cross-encoder.
03	Generate	Send the highest-ranked context to the answer model with an instruction to abstain when evidence is insufficient.
04	Inspect	Expose sources, diagnostic labels, timing, and structured telemetry for review.

Model role	Configured v1.0 model
Embedding	BAAI/bge-small-en-v1.5
Reranking	cross-encoder/ms-marco-MiniLM-L6-v2
Generation	openai/gpt-oss-120b
Rewriting and runtime diagnostics	openai/gpt-oss-20b

Live demonstration: <https://demo.measured-rag.com>

Retrieval that can be inspected and tuned

Measured RAG treats retrieval as a visible pipeline. Every retrieval route uses the same four stages, while route-specific weighting changes how dense and lexical evidence contribute to candidate ordering.

01	Dense retrieval	Embeddings identify semantically related chunks, including paraphrases that do not share exact wording.
02	BM25 retrieval	Lexical scoring rewards exact terms, names, and phrases that may be important to the question.
03	Weighted fusion	Dense and BM25 candidates are normalized and combined. Semantic, Keyword, and Balanced are weighted hybrid routes, not pure retrievers.
04	Cross-encoder reranking	A reranker scores the fused candidates jointly with the query and produces the final context order.

Routing modes

Mode	Purpose	Important boundary
Semantic	Favors dense similarity.	Still runs BM25 and fusion.
Keyword	Favors lexical matching.	Still runs dense retrieval and fusion.
Balanced	Uses a more even hybrid weighting.	The best balance depends on the document and question.
Auto	Selects one of the weighted routes from rewritten-query signals.	It is a fixed heuristic, not a learned confidence model.

Controls and tradeoffs

- Chunk size and overlap change the amount of local context and the chance that a semantic unit is split.
- Retrieval Top-N and the fusion threshold trade candidate recall against downstream work.
- Reranker candidate limits and Final Rerank Top-K trade context breadth against latency and prompt size.
- Query rewriting uses recent conversation history to resolve references, so earlier topics can affect a later query.
- Multi-document scope can restrict retrieval to selected files, but ambiguous selection can exclude intended evidence.
- Scores and thresholds control ranking. They are not calibrated probabilities of relevance or correctness.

Why this matters	A plausible answer can hide a weak retrieval path. By exposing chunks, routes, candidates, sources, and timing, Measured RAG makes it possible to distinguish retrieval behavior from answer fluency and to compare settings with evidence.
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Evidence, diagnostics, and hosted operation

The interface is designed for inspection, but inspection signals must be interpreted within their boundaries. Retrieved passages are ranked evidence supplied to the generator; they are not proof that the best passage was found or formal citation verification.

Indexed Chunks

Shows the extracted and chunked text available to retrieval. This is where layout-heavy or poorly parsed documents should be checked first.

Retrieved Sources

Shows the final passages supplied to generation, including source metadata and ranking signals.

Runtime diagnostics

Retrieval relevance and groundedness labels support operational review. They do not establish answer accuracy.

Hosted demo guardrails

Guardrail	Hosted value	Interpretation
Accepted formats	PDF, DOC, DOCX	Format acceptance does not guarantee faithful extraction.
Document allowance	3 documents	Shared-demo capacity boundary.
File and total upload	10 MiB each; 20 MiB total	Applies to accumulated uploaded files in the session.
Question length	2,000 characters	Cloud-specific request limit.
Questions per session	5 admitted questions	An admitted question can consume the allowance before later validation succeeds.

Privacy and lifecycle boundary

- Uploads are processed and indexed inside the running application container. Knowledge bases and conversations are session-scoped application state.
- Reset behavior manages visible application state but does not create a guaranteed deletion contract for temporary storage, platform systems, or external services.
- The shared demo has no confidentiality, account-based isolation, durable knowledge-base, or guaranteed-deletion contract. Do not upload confidential, regulated, personally identifiable, or otherwise sensitive documents.
- Hosted telemetry emits a constrained operational event without document text, questions, answers, filenames, source previews, diagnostic explanations, or conversation identifiers. Platform and provider logging remain separate concerns.

Local use is also not fully offline

Questions, recent conversation excerpts, retrieved passages, and generated answers are assembled into provider requests as needed for rewriting, generation, relevance checking, and groundedness checking. Provider handling is governed by the applicable external policy.

Recorded v1.0 evaluation

The approved publication evaluation uses a custom 28-paper, 129-question QASPER-derived subset. Each paper is indexed separately, and every question is asked with independent empty conversation history. The results describe this recorded condition only.

129 / 129 generation and semantic evaluation cases completed	23.78% normalized-token Answer F1	71.32% mean semantic correctness	84 / 129 semantic passes at a 0.70 threshold
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Median timing	Seconds	Interpretation
End-to-end answer	7.216	Recorded application answer path for the publication condition.
Retrieval total	5.266	Dense retrieval, BM25, fusion, reranking, and intervening retrieval work.
Reranking	5.158	Most recorded retrieval time was spent in reranking; this relationship can vary.

How to interpret the results

- Answer F1 measures normalized lexical overlap with references. Semantic judging measures reference-based compatibility. They measure different properties and are not combined.
- This is not an official full-split QASPER leaderboard result. The subset, configuration, models, and model-based judge bound the claim.
- Evidence F1 is not reported because retrieved chunks were not mapped to official QASPER evidence units.
- The semantic judge is model-based and nondeterministic. A diagnostic label or semantic pass is not a correctness guarantee.

Try these questions in Measured RAG
1. How does Measured RAG combine dense retrieval, BM25, fusion, and reranking?
2. What were the v1.0 benchmark results, and what limitations apply when interpreting them?
3. What privacy boundaries apply to the shared hosted demo?

Scope and attribution

Measured RAG v1.0 is optimized and evaluated primarily for English-language documents. Scanned or image-only PDFs have no OCR fallback; images, figures, tables, formulas, and complex layout relationships are not interpreted beyond extracted text.

Evaluation basis: Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. *A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers*. NAACL-HLT 2021, 4599-4610. [ACL Anthology](#). QASPER data is distributed under [CC BY 4.0](#).

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